

Validation examples of the Analytic Hierarchy Process and Analytic Network Process

Rozann Whitaker*

Creative Decisions Foundation, 4922 Ellsworth Avenue, Pittsburgh, PA, USA

Received 28 November 2006; accepted 14 March 2007

Abstract

One way to validate a scientific theory is to show that the results predicted by the theory give correct answers; that is, that they match known results. In the Analytic Hierarchy Process (AHP) this usually means finding examples with measures in an already known scale. To validate AHP priority vectors against measures from known scales the measures must first be normalized by dividing by their sum. When the two vectors are the same, or close, then one can say the results of the AHP model have been validated. The AHP and its generalization, the Analytic Network Process (ANP), can be validated at several levels ranging from priority vectors derived from pairwise comparison matrices to the synthesized priorities for a hierarchical model, to the priorities derived for the elements in an ANP network from the limiting supermatrix (perhaps most impressively validated by estimating market share of companies using intangible factors), to the overall results from complex ANP models involving several levels of networks. Many validation examples are presented along with a discussion of the compatibility index that can be used to measure closeness of priority vectors.

© 2007 Elsevier Ltd. All rights reserved.

Keywords: Validating the Analytic Hierarchy Process; Validating the Analytic Network Process; Compatibility index for the AHP; Validation examples for the AHP; Validation examples for the ANP

1. Introduction

“Scientific truth, that is to say the validity of an accepted theory, depends on two important kinds of factors: the guiding principles ... and what we have called the process of empirical verification ... these two factors are crucial in the establishment of any theory relating to any kind of knowledge”.

According to LeShan and Marganau [1], quoted above, scientific theories must be verified against things that happen in the real world and must follow the guiding principles of science that have been loosely enumerated as *simplicity, extensibility, multiple connections, logical fertility, stability of interpretation, causality, and elegance*.

In science validation is a central concern. It is as relevant in the social sciences, although more difficult to do, as it is in the physical sciences. Validating subjective observations made by experts or by experienced people with their

* Tel.: +1 412 621 6546.

E-mail address: rozann@creativedecisions.net.

perceptions is what we intend to deal with here. A decision theory that hides behind subjectivity and a claim to being normative (i.e., the theory specifies what is good for you) has no way of being validated even when it claims success. One cannot help but be struck by the rather precise way in which people using AHP are able to understand influences that they have internalized over a period of time. They can capture the importance of these influences by making pairwise comparison judgments expressed using the AHP fundamental scale in a matrix which results in a priority vector or in a hierarchical AHP model composed of a number of pairwise comparison matrices. Priority vectors can be compared to measures that have been transformed into relative form — that is, they have been normalized so that they sum to 1.

Rational decision-making is a talent we must encourage if we want to be more effective in implementing our ideas in the real world with its risks and resistance to change. There are two kinds of decisions. One is to determine what we prefer the most, known as *normative* decision making and the other is how to make a best choice given all the influences in the world around us that can affect the optimality of any choice we make, known as *descriptive* decision making. When we think about normative decisions it is easy to see why we do not wish anything to happen that can undermine the best choice we make and thus if we choose an alternative, we do not want it to be influenced by other alternatives that arrive (or occur to us) later. That is why preserving rank when a new alternative is added or an old one deleted is essential for this kind of thinking. Descriptive statements are falsifiable statements that attempt to describe the real world as it is. Normative statements legislate how things ought to be, what is good and what is bad; what is right and what is wrong. Normative statements can never be proven to be correct and workable, but only disproved with examples of what they recommended failing. A useful way to show their falsehood is by pointing to real life occurrences that they violate in spite of their claims.

But in reality how good any choice we make depends on how well we know our alternatives as compared with each other and with others outside the collection being compared so we can rank them as to how good they are. The drawback is that our knowledge of the alternatives may be very limited. It is known in practice that new alternatives can influence what we thought of in ranking other alternatives one by one earlier. Yet normative theories want to overlook that in all situations. A descriptive approach to decision making must cover both the normative and the descriptive ways of thinking. Such a theory must allow for enforcing the stability of choices in certain decisions and for their potential variability in others.

Decision-making involves prioritizing our ideas according to the circumstances we face now or might face in the future. A fundamental problem in decision-making is how to measure intangible criteria and how to interpret measurements of tangibles correctly so they can be combined with those of intangibles to yield sensible, not arbitrary numerical results. A crucial test is whether actual measurements can be used precisely as they are when needed.

The AHP/ANP is fundamentally a way to measure intangible factors by using pairwise comparisons with judgments that represent the dominance of one element over another with respect to a property that they share. Many examples are worked out by knowledgeable people without entering all the judgments but only contrasting ones that form a spanning tree which covers all the elements thus shortening the time in which the exercise is done. The AHP/ANP has found useful applications in decision making which involves numerous intangibles. It is a process of laying out a structure of all the essential factors that influence the outcome of a decision. Numerical pairwise comparison judgments are then elicited to express people's understanding of the importance, preference or likely influence of these elements on the final outcome obtained by synthesizing the priorities derived from different sets of pairwise comparisons. Sensitivity analysis is performed in the end to determine the stability of the outcome to wide perturbations in the judgments. The process has been validated in practice in many ways as we show in the next section.

The AHP uses additive synthesis which means multiplying the priorities of the alternatives by those of their respective criteria and adding. It was once proposed by F. Lootsma of Delft that it would be better to use multiplicative synthesis by raising the alternative priorities to the power of their respective criterion priorities and multiplying the resulting priorities of the alternatives. He had a few advocates who published on it, but no one has promoted this idea since T. Saaty brought to the attention of Lootsma and others several years ago that an alternative with the same priority (value less than one) under two criteria when raised to the respective powers of the criteria would result in the alternative under the larger priority criterion receiving a smaller value — contradicting common sense. Consider for example raising the fraction $1/2$ to the power of .7 (which yields .6156) and then to the power of .3 (which yields .8123). Because of this multiplicative synthesis is not a viable option for synthesis.

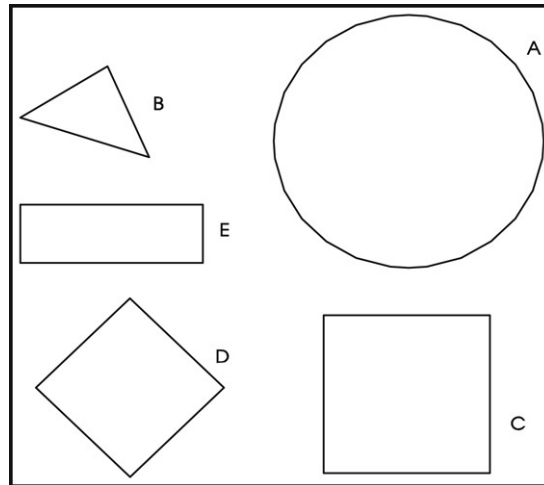


Fig. 1. Comparison of five areas.

2. Single matrix validation examples

In this section we illustrate the validation of the AHP/ANP with a variety of example for pairwise comparison matrices, hierarchies and networks. Because the process deals with the idea of dominance in the forms of importance, preference and likelihood, when importance is applied to the criteria and preference to the alternatives we have a normative outcome which as we said above has no way to defend its validity except for counter examples in real life. For that we mention a statement made by Buede and Maxwell [2] who performed many experiments to compare the outcomes of different methods and used their preferred approach known as Multi-attribute Value Theory (MAVT) as the basis for the comparisons of the methods. They wrote: *“These experiments demonstrated that the MAVT and AHP techniques, when provided with the same decision outcome data, very often identify the same alternatives as ‘best’. The other techniques are noticeably less consistent with MAVT, the Fuzzy algorithm being the least consistent”*.

In its descriptive/predictive form the AHP generally uses importance for the dominance among the criteria and likelihood for dominance among the alternatives.

Relative sizes of areas

Here is a validation exercise we have often used with individuals when introducing them to the Analytic Hierarchy Process. We ask the individual to estimate the relative areas of the five geometric shapes shown in Fig. 1 using the AHP. The person makes pairwise comparison judgments as to how much bigger in area the first shape is than the second as shown in the matrix in Table 1. For example, an area of 12 square units is 3 times bigger than an area of 4 square units. So the judgment 3 is used. Use the numbers 1 to 9 (or their reciprocals if the second of the pair is bigger than the first) from the Fundamental Scale of the AHP; decimals in between are allowed. The judgments are entered in the upper triangular half of the matrix. The entire exercise requires the individual to make $n(n-1)/2$ judgments, where n is the number of things being compared, in this case $n = 5$, so for this exercise 10 judgments are required. Enter the reciprocals of those judgments in the bottom triangular half of the matrix. The relative sizes of the figures are then given by the principal eigenvector of the matrix or, in the absence of a computer, approximately by normalizing each column to one and taking the average of corresponding entries in the columns. This is the priority vector of relative values. It is remarkable how close one can come to the actual relative sizes in this way. Often verbal judgments are made first particularly when dealing with intangibles.

The actual relative values of these areas are approximately $A = 0.47$, $B = 0.05$, $C = 0.24$, $D = 0.14$, and $E = 0.09$. Here A is 47% of the overall area of the five figures, B is 5% and so on. Or, to put it another way, $A/B = 0.47/0.05 = 9.4$, and A is about 9 times as large as B , and the relative sizes of the other figures can be determined in the same way.

If there is no computer available, a shortcut for computing the principal eigenvector is to assume the matrix is consistent, normalize each column and then take the average of the corresponding entries in the columns to obtain the

Table 1
One individual's judgment matrix

	Circle	Triangle	Square	Diamond	Rectangle	Eigenvector (priority vector of relative sizes)
Circle	1	9	2.5	3	6	0.488
Triangle	1/9	1	1/5	1/3.5	1/1.5	0.049
Square	1/2.5	5	1	1.7	3	0.233
Diamond	1/3	3.5	1/1.7	1	1.5	0.148
Rectangle	1/6	1.5	1/3	1/1.5	1	0.082

Only the judgments in bold are made, the others are automatically determined.

Table 2
Pairwise comparisons of the weights of five objects

Weight	Radio	Typewriter	Large attache case	Projector	Small attache case	Eigenvector	Actual relative weights
Radio	1	1/5	1/3	1/4	4	.09	.10
Typewriter	5	1	2	2	8	.40	.39
Large attache case	3	1/2	1	1/2	4	.18	.20
Projector	4	1/2	2	1	7	.29	.27
Small attache case	1/4	1/8	1/4	1/7	1	.04	.04

Table 3
Relative electricity consumption (kilowatt hours) of household appliances

Annual electric consumption	Elec. range	Refrig.	TV	Dish wash.	Iron	Radio	Hairdryer	Eigenvector	Actual relative weights
Electric range	1	2	5	8	7	9	9	.393	.392
Refrigerator	1/2	1	4	5	5	7	9	.261	.242
TV	1/5	1/4	1	2	5	6	8	.131	.167
Dishwasher	1/8	1/5	1/2	1	4	9	9	.110	.120
Iron	1/7	1/5	1/5	1/4	1	5	9	.061	.047
Radio	1/9	1/7	1/6	1/9	1/5	1	5	.028	.028
Hairdryer	1/9	1/9	1/8	1/9	1/9	1/5	1	.016	.003

priority vector. The priority vector shown in the right column of the matrix is the actual eigenvalue solution, but using the shortcut calculation gives a very similar result: $A = 0.486$, $B = 0.049$, $C = 0.233$, $D = 0.148$, $E = 0.082$.

It is easy to make consistent judgments in this exercise because we are dealing with objects with measures that are quite familiar to most people. If one were to measure the figures, compute the ratios of their areas exactly and enter these ratios instead of using judgments, the priority vector would give back the exact relative areas of the figures. The point here is to show that judgments can give accurate results. Sometimes people try to measure the figures and use formulas dimly remembered from high school geometry to verify the results. The errors associated with physically measuring the figures often give worse results than eyeballing the figures and making judgments.

An interesting thing often happens when the exercise is tried with a group. In the Analytic Hierarchy Process the judgments of the members of the group on a pair are combined using the geometric mean. When the combined judgments are used in the matrix often the answer is closer to the actual relative sizes of the figures than most of the individuals' answers are.

Relative weights of objects

Table 2 gives the estimated pairwise comparisons of the weights of the five objects lifted by hand, made by the then President of the Diners Club, a friend of the author. The two vectors appear to be very close but are they really close? To determine closeness of two priority vectors one must use the Saaty Compatibility Index.

Relative electric consumption of household appliances

In Table 3 we have paired comparisons done by students in Electrical Engineering estimating the consumption of electricity of common household appliances. How compatible are the derived and actual vectors?

Table 4

Paired comparisons of the relative wealth of seven nations

	US	USSR	China	France	UK	Japan	W. Germany
US	1	4	9	6	6	5	5
USSR	1/4	1	7	5	5	3	4
China	1/9	1/7	1	1/5	1/5	1/7	1/5
France	1/6	1/5	5	1	1	1/3	1/3
UK	1/6	1/5	5	1	1	1/3	1/3
Japan	1/5	1/3	7	3	3	1	2
W. Germany	1/5	1/4	5	3	3	1/2	1

Table 5

The relative priorities compared with the actual values normalized

	Normalized eigenvector	Actual GNP (1972)	Normalized GNP values
US	.427	1167	.413
USSR	.23	635	.225
China	.021	120	.043
France	.052	196	.069
UK	.052	154	.055
Japan	.123	294	.104
W. Germany	.094	257	.091

The hairdryer is of such a small magnitude that it probably should have been left out to make the set homogeneous.

Relative wealth of seven nations

The pairwise comparison matrix shown in Table 4 to estimate the relative wealth of seven leading nations was done on an airplane in 1973 by Thomas Saaty and Mohammad Khouja [3] by simply using their common knowledge about the relative power and standing of these countries in the world and without referring to any specific economic data related to GNP values. The relative priorities derived from the pairwise comparison matrix are shown in the first column of Table 5 and the actual relative values are given in the last column.

3. Hierarchic validation examples

To make good applications needs expert knowledge of the subject, a structure that represents the pertinent issues, and a little time to do justice to the subject. In this part we give three hierarchic examples that gave results close to what the values actually were and all the works were published in refereed journals.

World chess championship outcome validation — Karpov–Korchnoi match

The following criteria and hierarchy were used to predict the outcome of world chess championship matches using judgments of ten grandmasters in the then Soviet Union and the United States who responded to questionnaires they were mailed [4]. The predicted outcomes that included the number of games played, drawn and won by each player either was exactly as they turned out later or adequately close to predict the winner. The outcome of this exercise was notarized before the match took place. The notarized statement was mailed to the editor of the *Journal of Behavioral Sciences* 1980, along with the paper later. See the book by Saaty and Vargas on prediction [5], Projection and Forecasting, Kluwer, 1991. The prediction was that Karpov would win by 6 to 5 games over Korchnoi, which he did. In Table 6 we have the different criteria or factors used to judge the winner by world chess masters in reply to a questionnaire. Fig. 2 represents the layout of the decision.

Monetary exchange rate — dollar versus the yen

In the late 1980's three economists at the University of Pittsburgh, Professors A. Blair, R. Nachtmann, and J. Olson, worked with Thomas Saaty [6] on predicting the yen/dollar exchange rate. The predicted value was fairly close to the

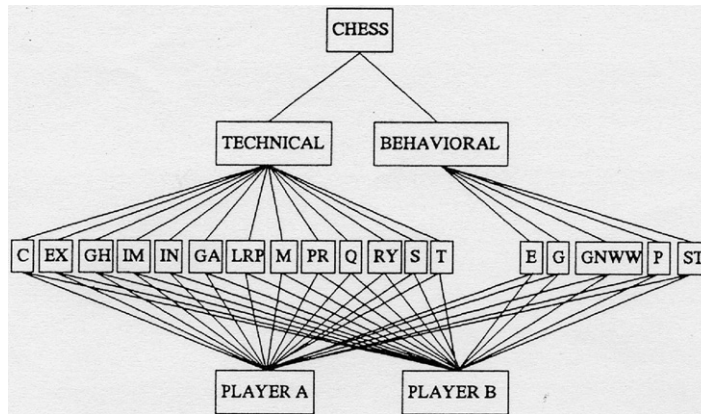


Fig. 2. The chess win prediction hierarchy.

Table 6
Definitions of chess factors

T (1)	Calculation (Q): The ability of a player to evaluate different alternatives or strategies in light of prevailing situations.
B (2)	Ego (E): The image a player has of himself as to his general abilities and qualification and his desire to win.
T (3)	Experience (EX): A composite of the versatility of opponents faced before, the strength of the tournaments participated in, and the time of exposure to a rich variety of chess players.
B (4)	Gamesmanship (G): The capability of a player to influence his opponent's game by destroying his concentration and self-confidence.
T (5)	Good Health (GH): Physical and mental strength to withstand pressure and provide endurance.
B (6)	Good Nerves and Will to Win (GN): The attitude of steadfastness that ensures a player's health perspective while the going gets tough. He keeps in mind that the situation involves two people and that if he holds out the tide may go in his favor.
T (7)	Imagination (IW): Ability to perceive and improvise good tactics and strategies.
T (8)	Intuition (IN): Ability to guess the opponent's intentions.
T (9)	Game Aggressiveness (GA): The ability to exploit the opponent's weaknesses and mistakes to one's advantage. Occasionally referred to as "killer instinct".
T (10)	Long Range Planning (LRP): The ability of a player to foresee the outcome of a certain move, set up desired situations that are more favorable, and work to alter the outcome.
T (11)	Memory M: Ability to remember previous games.
B (12)	Personality (P): Manners and emotional strength, and their effects on the opponent in playing the game and on the player in keeping his wits.
T (13)	Preparation (PR): Study and review of previous games and ideas.
T (14)	Quickness (Q): The ability of a player to see clearly the heart of a complex problem.
T (15)	Relative Youth (RY): The vigor, aggressiveness, and daring to try new ideas and situations, a quality usually attributed to young age.
T (16)	Seconds (S): The ability of other experts to help one to analyze strategies between games.
B (17)	Stamina (ST): Physical and psychological ability of a player to endure fatigue and pressure.
T (18)	Technique M: Ability to use and respond to different openings, improvise middle game tactics, and steer the game to a familiar ground to one's advantage.

average value for a considerable number of months after that. Fig. 3 represents the structure of the decision and gives the outcome at the bottom.

Number of children in rural Indian families

In a hierarchy whose goal is the optimal family size in India (from a study published in the Journal of Mathematical Sociology, 1983, Vol. 9 pp. 181–209, there were four major criteria of Culture (with subcriteria: Religion, Women Status, Manhood), Economic factors (with subcriteria: Cost of child Rearing, Old Age security, Labor, Economic Improvement, Prestige and Strength), Demographic factors (with subcriteria: Short Life Expectancy, High Infant Mortality) and the Availability and acceptance of Contraception (with subcriteria: High Level of Availability and Acceptance of contraception, Medium level of Availability and Acceptance of contraception, low level of Availability and Acceptance of contraception. At the bottom three alternatives were considered: Families with 3 or Less Children,

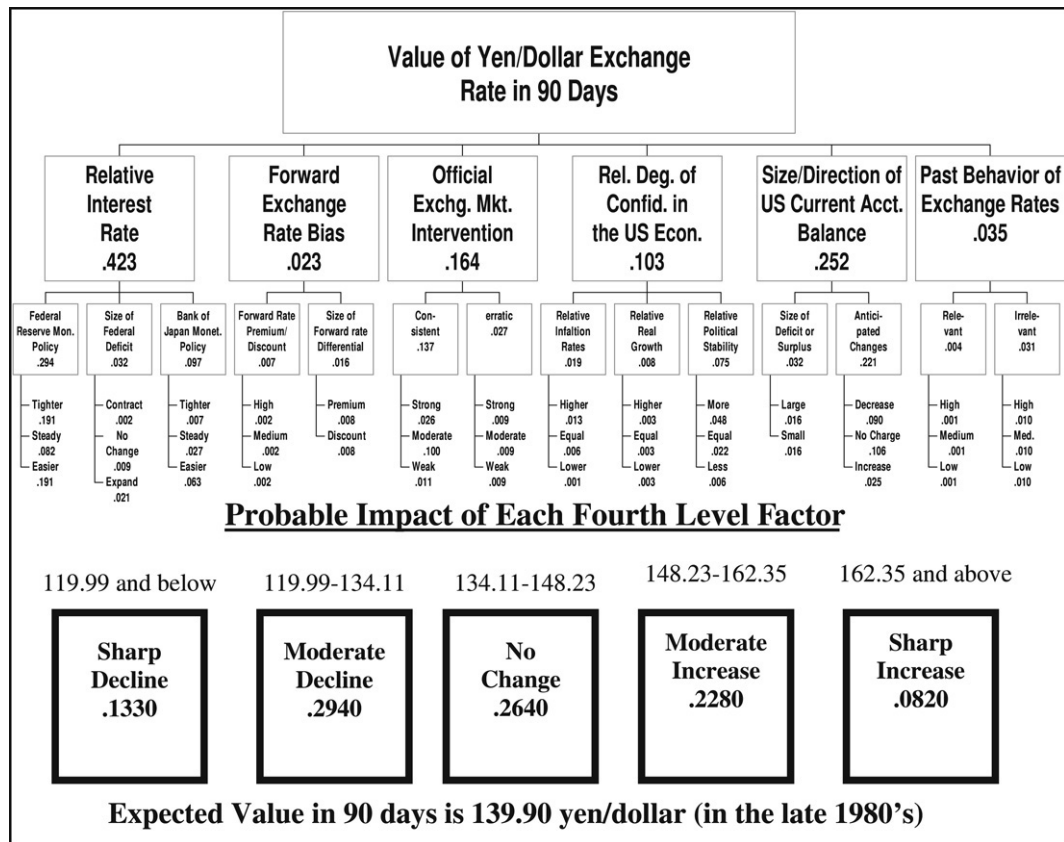


Fig. 3. Predicting the yen/dollar exchange rate.

Families with 4 to 7 Children, and Families with 8 or More Children. The outcome of this example for reasons explained in the research paper had two projections of 5.6 and 6.5 children per family (due to regional differences.) The actual value we obtained from the literature after the study was done were 6.8 births per woman in 1972 and 5.6 in 1978.

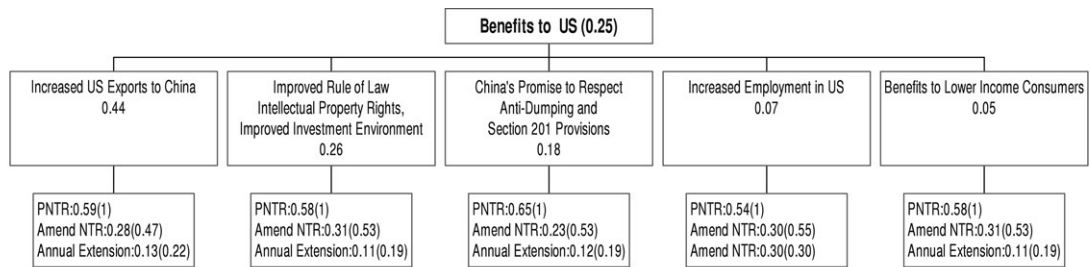
Decision by the US Congress on backing China to join the World Trade Organization (WTO)

Saaty and Cho [7] did this study in 1999. In May of the year 2000 the US Congress passed a bill to grant China Permanent Normal Trade Relations status with no conditions attached and soon after China was accepted as a member of the WTO.

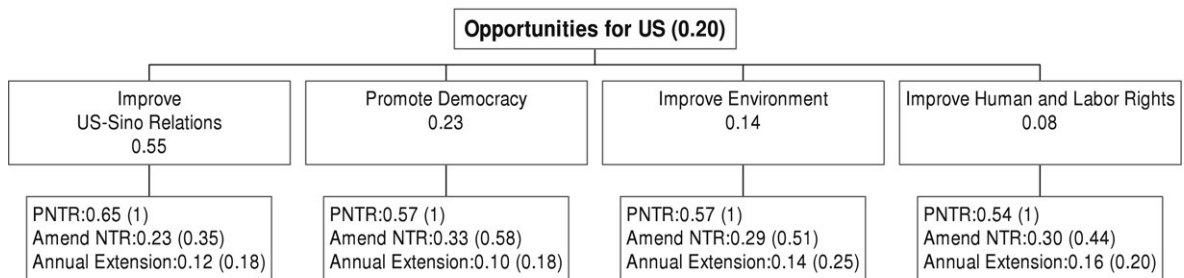
Briefly, the alternatives of the decision are:

1. *Passage of a clean PNTR bill:* Congress grants China Permanent Normal Trade Relations status with no conditions attached. This option would allow implementation of the November 1999 WTO trade deal between China and the Clinton administration. China would also carry out other WTO principles and trade conditions.
2. *Amendment of the current NTR status bill:* This option would give China the same trade position as other countries and disassociate trade from other issues. As a supplement, a separate bill may be enacted to address other matters, such as human rights, labor rights, and environmental issues.
3. *Annual extension of NTR status:* Congress extends China's Normal Trade Relations status for one more year, and, thus, maintains the status quo.

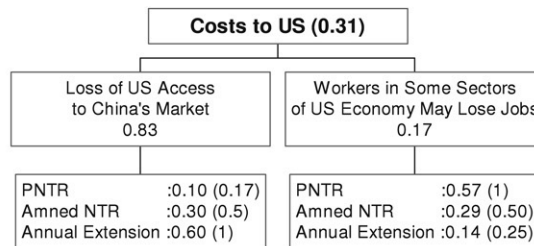
The four hierarchies shown in Fig. 4 were used and their outcomes were combined to derive the final priorities that show how Congress was going to vote. In fact China was later admitted to the WTO as was recommended here. Figs. 4 and 5 summarize the process and the results.



Benefits Synthesis (Idealized form): PNTR 1.00, Amend NTR 0.51, Annual Extension 0.21



Opportunities Synthesis (Idealized form): PNTR 1, Amend NTR 0.43, Annual Extension 0.13



Costs Synthesis (which is more costly, Idealized form): PNTR 0.31, Amend NTR 0.50, Annual Extension 0.87



Risks Synthesis (which is more risky, Idealized form): PNTR 0.54, Amend NTR 0.53, Annual Extension 0.58

Fig. 4. Hierarchies for evaluating alternatives with respect to benefits, costs, opportunities, and risks.

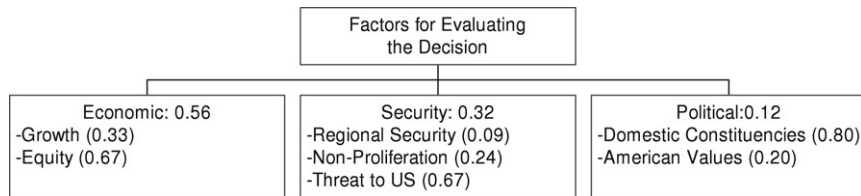


Fig. 5. Prioritizing the strategic criteria to be used in rating the BOCR.

Table 7

Priority ratings for the merits: benefits, costs, opportunities, and risks

		Benefits	Opportunities	Costs	Risks
Economic (0.56)	Growth (0.19)	High	Medium	Very low	Very low
	Equity (0.37)	Medium	Low	High	Low
Security (0.32)	Regional (0.03)	Low	Medium	Medium	High
	Non-Proliferation (0.08)	Medium	High	Medium	High
	Threat to US (0.21)	High	High	Very high	Very high
Political (0.12)	Constituencies (0.1)	High	Medium	Very high	High
	American values (0.02)	Very low	Low	Low	Medium
Priorities		0.25	0.20	0.31	0.24

Intensities: Very High (0.42), High (0.26), Medium (0.16), Low (0.1), Very Low (0.06).

Table 8

Two methods of synthesizing results from the BOCR hierarchies using the ideal model

Alternatives	Benefits (0.25)	Opportunities (0.20)	Costs (0.31)	Risks (0.24)	BO/CR	$bB + oO - cC - rR$
PNTR	1	1	0.31	0.54	5.97	0.22
Amend NTR	0.51	0.43	0.50	0.53	0.83	−0.07
Annual exten.	0.21	0.13	0.87	0.58	0.05	−0.33

How to derive the priority shown next to the goal of each of the four hierarchies in Fig. 4 is outlined in Table 7. We rated each of the four merits: benefits, costs, opportunities and risks of the dominant PNTR alternative, as it happens to be in this case, in terms of intensities for each assessment criterion. The intensities, Very High, High, Medium, Low, and Very Low were themselves prioritized in the usual pairwise comparison matrix to determine their priorities. We then assigned the appropriate intensity for each merit on all assessment criteria. The outcome is as found at the bottom row of Table 7.

We are now able to obtain the overall priorities of the three major decision alternatives listed earlier, given as columns in Table 8 which gives three ways of synthesizing for the ideal mode, we see in bold that PNTR is the dominant alternative any way we synthesize as in the last two columns of Table 8.

4. Network validation examples

4.1. Turn around of the US economy

Let us consider the problem of the turn around of the US economy and introduce 3, 6, 12, 24 month time periods at the bottom [10]. Decomposing the problem hierarchically, the top level consists of the primary factors that represent the forces or major influences driving the economy: “Aggregate Demand” factors, “Aggregate Supply” factors, and “Geopolitical Context”. Each of these primary categories was then decomposed into subfactors represented in the second level. Under Aggregate Demand, we identified consumer spending, exports, business capital investment, shifts in consumer and business investment confidence, fiscal policy, monetary policy, and expectations with regard to such

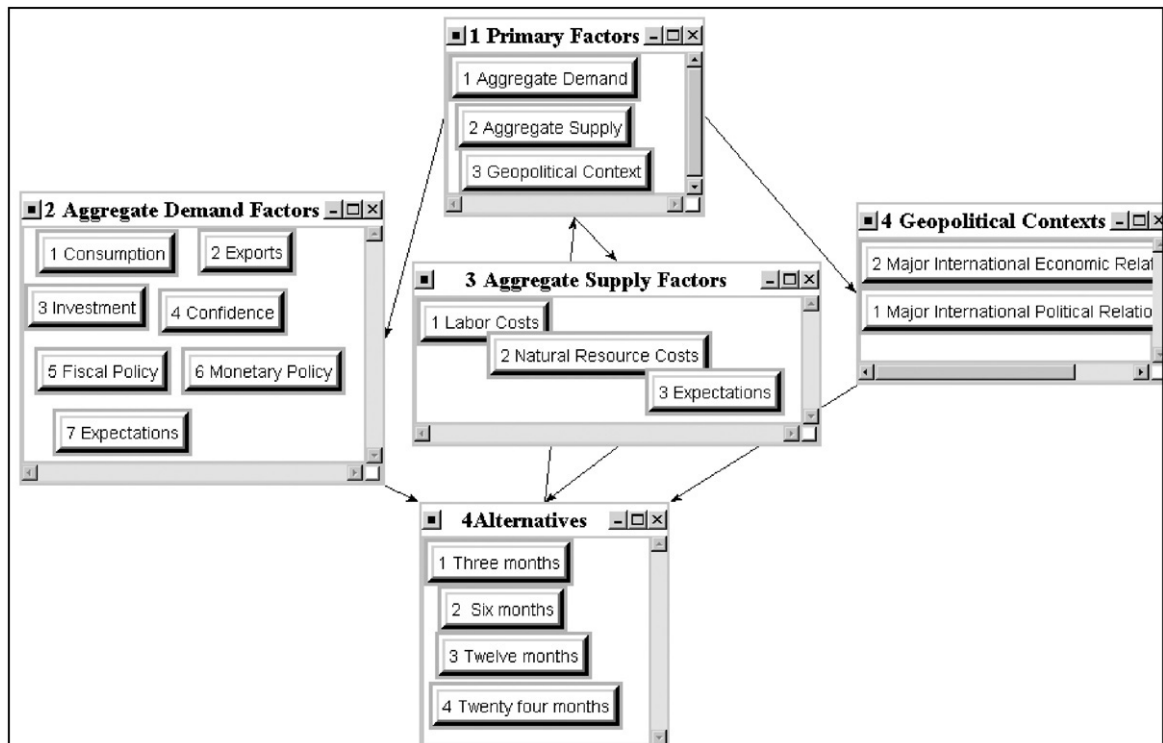


Fig. 6. Overall view of the year 2001 economic prediction model.

questions as the future course of inflation, monetary policy and fiscal policy. (We make a distinction between consumer and business investment confidence shifts and the formation of expectations regarding future economic developments.)

Under Aggregate Supply, we identified labor costs (driven by changes in such underlying factors as labor productivity and real wages), natural resource costs (e.g., energy costs), and expectations regarding such costs in the future. With regard to Geopolitical Context, we identified the likelihood of changes in major international political relationships and major international economic relationships as the principal subfactors. With regard to the subfactors under Aggregate Demand and Aggregate Supply, we recognized that they are, in some instances, interdependent. For example, a lowering of interest rates as the result of a monetary policy decision by the Federal Reserve should induce portfolio rebalancing throughout the economy. In turn, this should reduce the cost of capital to firms and stimulate investment, and simultaneously reduce financial costs to households and increase their disposable incomes. Any resulting increase in disposable income stimulates consumption and, at the margin, has a positive impact on employment and GNP. This assumes that the linkages of the economy are in place and are well understood. This is what the conventional macroeconomic conceptual models are designed to convey.

The third level of the hierarchy consists of the alternate time periods in which the resurgence might occur as of April 7, 2001: within three months, within six months, within twelve months, and within twenty-four months. Because the primary factors and associated subfactors are time dependent, their relative importance had to be established in terms of each of the four alternative time periods. Thus, instead of establishing a single goal as one does for a conventional hierarchy, we used the bottom level time periods to compare the two factors at the top. This entailed creation of a feedback hierarchy known as a “holarchy” in which the priorities of the elements at the top level are determined in terms of the elements at the bottom level, thus creating an interactive loop. Fig. 6 provides a schematic representation of the holarchy we used to forecast the timing of the economic resurgence.

To obtain our forecast in terms of months, we multiplied each priority by the mid-point of its corresponding time interval and added the results (as one does when evaluating expected values) as shown in Table 9.

We interpret this to mean that the recovery would occur 8.54 months from the time of the forecasting exercise, or in the fall. The Wall Street Journal of July 18, 2003, more than two year after the exercise had the following to say about the turnaround date as shown in Fig. 7.

Table 9
Summary of results of the forecast

Time period	Mid-point of time period	Priority of time period	Mid-point $\times$ priority
	(Expressed in months from present, with the current month as 0.)		
Three months	$0 + (3 - 0)/2 = 1.5$	0.30581	0.45871
Six months	$3 + (6 - 3)/2 = 4.5$	0.20583	0.92623
Twelve months	$6 + (12 - 6)/2 = 9.0$	0.18181	1.63629
Twenty-four months	$12 + (24 - 12)/2 = 18.0$	0.30656	5.51808
Total			8.53932

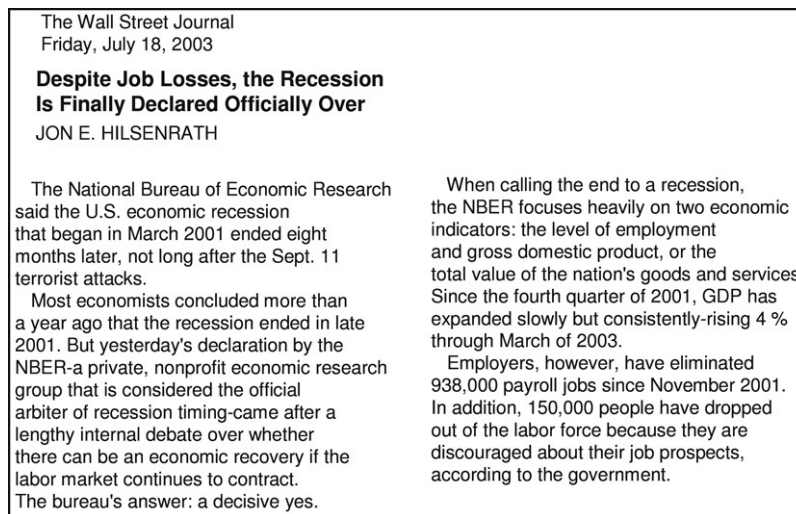


Fig. 7. The wall street journal on when the recession was over.

Market share for the cereal industry in 2002

The following is one of numerous validation examples done by graduate students in Thomas L. Saaty's Complex Decision Making class. These are students getting their MBA degrees and for the most part they work full time during the day at some company. Many of the market share estimation examples are done in class in about one hour and without access to data. The students look up the validating data later on the Internet. The example below was developed by Stephanie Gier and Florian John in March, 2002. They write: To become familiar with the Super Decision software we have chosen to estimate the market shares for the Ready-to-Eat-Breakfast-Cereal industry. This idea was born after eating a delicious breakfast of Post's OREO O's cereal. To see how good our assumptions were, we compared our calculated results with the latest market share data available for the competing companies for 2001. First we created the model. We identified 6 major competitors in the ready to eat cereal market, Kellogg, General Mills, Post, Quaker, Nabisco and Ralston as our alternatives. There were more companies in this market having an actual cumulative market share of roughly about 6% that it turned out later that we had left out. Since we were only concerned with deriving relative values, the relative shares of other residual companies do not matter.

Major impacts on the companies' market shares are:

- Price of the products offered (named cost for the consumer).
- Advertising/sales ratio (how much money is spend for advertising).
- Shelf space (places where the products are located in the stores).
- Tools (selling tools used to increase sales and market shares).
- Distribution/availability (major distribution channels used to sell the product).

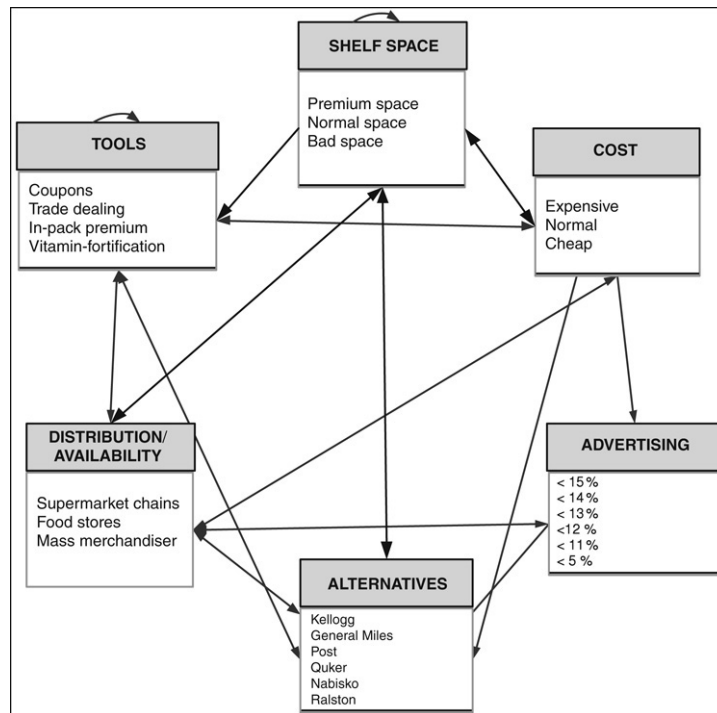


Fig. 8. Cereal manufacturers market share.

Table 10
Priorities derived from the model versus the actual income data (normalized)

Alternatives	Kellogg	General mills	Post	Quaker	Nabisco	Ralston
Model priorities	0.324	0.255	0.147	0.116	0.071	0.087
Actual	0.342	0.253	0.154	0.121	0.057	0.073

These five major impacts (clusters) are further divided in the following nodes:

Tools: (Coupons, trade dealing, in-pack premiums, vitamin fortifications).

Distribution: (Supermarket Chains, Food Stores, Mass Merchandiser).

Shelf Space: (Premium Space, Normal Space, Bad Space).

Cost: (Expensive, Normal, Cheap).

Advertising: (< 15%, < 14%, < 13%, < 12%, < 11%, < 5%).

The links are depicted in the model shown in Fig. 8. The final result is given in Table 10 in which we compare the outcome from the model with the normalized actual values. Third we compared our calculated market shares with the real market share data for 2001. The table that follows lists estimated market share values and the actual ones taken from the Website of the International Data Corporation.

The compatibility index value of 1.01403 (very good) is obtained by multiplying elementwise the matrix of pairwise ratios of the model estimate data, by the transpose of the matrix of pairwise ratios of the actual normalized data, adding all the resulting entries and dividing by n^2 . The compatibility index should not be more than 1.1.

Let us describe the calculations needed to derive the result in the “Model Estimate” row of the table. From the pairwise comparison judgments we constructed a super matrix, from the various sets of pairwise comparisons that were required using the ANP software SuperDecisions. Then we weighted blocks of the supermatrix by the corresponding entries from the matrix of priority vectors of paired comparisons of the influence of all the clusters on each cluster with respect to market share whose derived priorities for each matrix are represented in a column of Table 11. This yielded the weighted supermatrix that is now stochastic as its columns add to one. We raised this matrix to limiting powers to obtain the overall model priorities given in Table 10.

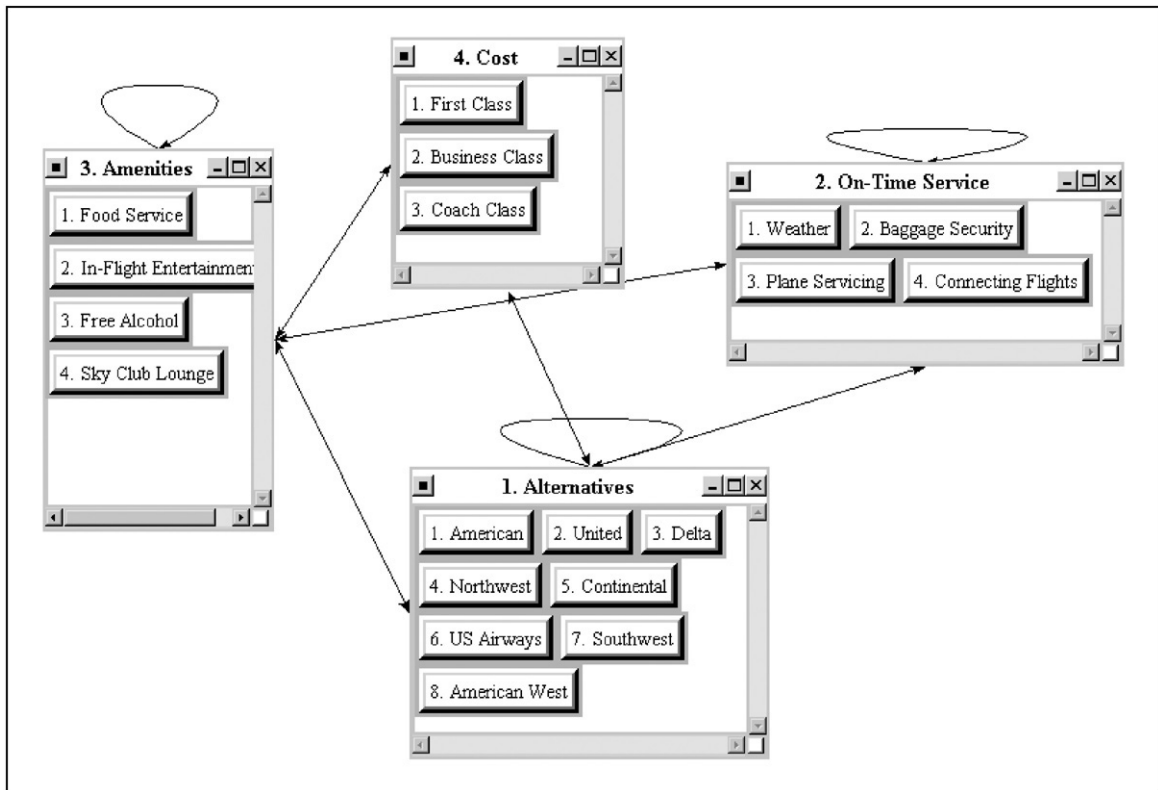


Fig. 9. Airline market share model using the ANP SuperDecisions software.

Table 11
Cluster matrix priorities

	Advertising	Alternatives	Cost	Distrib./Availability	Shelf space	Tools
Advertising	0.000	0.184	0.451	0.459	0.000	0.000
Alternatives	0.000	0.000	0.052	0.241	0.192	0.302
Cost	0.000	0.575	0.000	0.064	0.044	0.445
Distribution/availability	0.000	0.107	0.089	0.000	0.364	0.159
Shelf space	0.000	0.071	0.107	0.084	0.297	0.000
Tools	0.000	0.062	0.302	0.152	0.103	0.095

Market share for the airline industry in 2001

James Nagy did the following study of the market share of eight US airlines in 2001. Comparative income data was not available yet for 2001, so data for the year 2000 was used. Nowhere did he use numerical data, but only his knowledge of the airlines and how good each is relative to the others on the factors mentioned below. Note that in three of the clusters there is an inner dependence loop that indicates that the elements in that cluster depend on each other with respect to market share. Table 12 gives the final estimated and the actual relative values that are again very close. Fig. 9 shows the network model of clusters and their inner and outer dependence connections that produced these results. Nagy writes: “I initially chose the airline industry for the assignment because I was a frequent traveler. My study group at Katz helped me make the comparisons between airlines that I did not have first hand experience on as a passenger. Otherwise, I used my personal experience and perception of consumer sentiment towards the airlines to make the comparison. I was equally surprised at the results. In fact, I initially questioned how they could be so close. I would like to see the results of a study using today’s consumer perception. A lot has changed in the industry since the 9/11 tragedy in the year 2001. You could divide the class up into 4 to 5 small groups and let them do the comparisons as individual groups and compare the results”.

Table 12
Market share of airlines, actual and predicted

	Actual (yr 2000)	Model estimate
American	23.9	24.0
United	18.7	19.7
Delta	18.0	18.0
Northwest	11.4	12.4
Continental	9.3	10.0
US Airways	7.5	7.1
Southwest	5.9	6.4
American West	4.4	2.9

The adjacency and path matrices

The question often arises as to why raising the matrix of judgments or sometimes the supermatrix to a power determines for its entries the number of paths between the corresponding vertices, whose length is equal to that power. The vertex matrix is a first step for understanding this concept. We define a vertex (or adjacency) matrix for both directed and undirected graphs. The element in the (i, j) position of the matrix is equal to the number of edges incident with both vertex i and vertex j (or directed from vertex i to vertex j in the directed case). Thus, for the directed graph of a network we might have:

$$\vec{V} = \begin{matrix} & v_1 & v_2 & v_3 & v_4 & v_5 & v_6 \\ \begin{matrix} v_1 \\ v_2 \\ v_3 \\ v_4 \\ v_5 \\ v_6 \end{matrix} & \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 1 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \end{matrix}.$$

In general, we have the following theorem regarding the vertex matrix $\vec{V}$ of a graph.

Theorem. The matrix $\vec{V}^n$ gives the number of arc progressions of length n between any two vertices of a directed graph.

Proof. If a_{ik} is the number of arcs joining v_i to v_k and a_{kj} is the number of arcs joining v_k to v_j , then $a_{ik}a_{kj}$ is the number of different paths each consisting of two arcs joining v_i to v_j and passing through v_k . If this is summed over all values of k , that is, over all the intermediate vertices, one obtains the number of paths of length 2 between v_i and v_j . If we now use a_{ij} to form $a_{ij}a_{jm}$, we have the number of different paths of length 3 between v_i and v_m passing through v_j , and so on. Thus if we assume the theorem is true for $\vec{V}^{n-1}$, then the coefficients of $\vec{V}^n = \vec{V}^{n-1}\vec{V}$ give the number of paths of length n between corresponding vertices. This completes the proof. A similar theorem holds for undirected graphs.

Relationship between the supermatrix of the ANP and input–output econometric analysis

The following proof of the relationship between the supermatrix of the ANP and Leontieff's Input–Output Model was given by Vargas [8]. Fig. 10 depicts interdependence in economic input–output models.

Let $\mathbf{A}$ be the matrix of relative input–output coefficients. Let $\mathbf{v}$ and $\mathbf{d}$ be the relative value added and the relative final demand vectors. Let x_{ij} the amount of resource that sector j receives from sector i . Let v_j be the value added corresponding to sector j and let d_i be the final demand of sector i . The value added of a sector includes wages and salaries, profit-type income, interest, dividends, rents, royalties, capital consumption allowances and taxes. The final demand of a sector includes imports, exports, government purchases, changes in inventory, private investment and

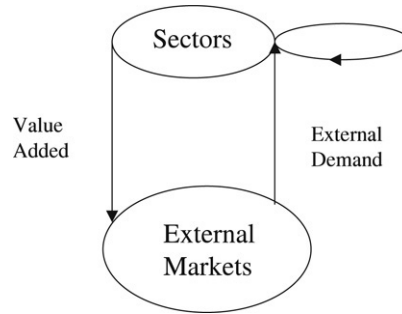


Fig. 10. Input–output network.

sometimes, household purchases. Thus, the input–output matrix is given by:

$$\begin{array}{c} \text{Sectors} \\ S_1 \\ S_2 \\ \vdots \\ S_n \end{array} \quad \begin{array}{ccccc} S_1 & S_2 & \cdots & S_n & \\ \left(\begin{array}{ccccc} x_{11} & x_{12} & \cdots & x_{1n} & d_1 \\ x_{21} & x_{22} & \cdots & x_{2n} & d_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{nn} & d_n \\ v_1 & v_2 & \cdots & v_n & 0 \end{array} \right) \end{array}.$$

Let $I_1, I_2, \dots, I_n$ be the total input to the sectors and let $O_1, O_2, \dots, O_n$ be the total output of the sectors, i.e., $\sum_{j=1}^n x_{ij} + d_i = O_i$ and $\sum_{i=1}^n x_{ij} + v_j = I_j$. Let the relative input–output coefficients be given by $a_{ij} \equiv \frac{x_{ij}}{I_j}$. The relative final demand of a sector with respect to the other sectors is given by $a_{i,n+1} \equiv \frac{d_i}{\sum_{h=1}^n d_h}$. On the other hand, the relative value added is given by $a_{n+1,1} \equiv \frac{v_1}{I_1}$. Thus, the matrix of interactions represented by the network in Fig. 9 is given by:

$$W = \begin{pmatrix} \mathbf{A} & \bar{\mathbf{a}}_{\bullet,n+1} \\ \bar{\mathbf{a}}_{n+1,\bullet} & 0 \end{pmatrix} = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} & a_{1,n+1} \\ a_{21} & a_{22} & \cdots & a_{2n} & a_{2,n+1} \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} & a_{n,n+1} \\ a_{n+1,1} & a_{n+1,2} & \cdots & a_{n+1,n} & 0 \end{pmatrix}.$$

Because W is a stochastic irreducible matrix, $\lim_{k \rightarrow \infty} W^k$ is given by $\bar{w}e^T$ where $\bar{w}$ is a $(n+1) \times 1$ vector that is the principal right eigenvector of W , and $e^T = (1, 1, \dots, 1)$ is the $1 \times (n+1)$ unit vector. Thus, we have $W\bar{w} = \bar{w}$ or

$$\begin{aligned} \sum_{j=1}^n a_{ij} w_j + w_{n+1} a_{i,n+1} &= w_i, \quad i = 1, 2, \dots, n \\ \sum_{j=1}^n a_{n+1,j} w_j &= w_{n+1}. \end{aligned}$$

In matrix notation we have:

$$\begin{aligned} (I - A)\bar{w}_n &= w_{n+1}\bar{\mathbf{a}}_{\bullet,n+1} \\ \mathbf{a}_{n+1,\bullet}\bar{w}_n &= w_{n+1} \end{aligned}$$

where $\bar{w}_n = (w_1, \dots, w_n)^T$. Thus, we have $\bar{w}_n = w_{n+1}(I - A)^{-1}\bar{\mathbf{a}}_{\bullet,n+1}$ and hence, we can write

$$\lim_{k \rightarrow \infty} W^k = \begin{pmatrix} w_{n+1}(I - A)^{-1}\bar{\mathbf{a}}_{\bullet,n+1} \\ w_{n+1} \end{pmatrix} e^T.$$

Note that $\bar{w}_n = w_{n+1}(I - A)^{-1}\bar{\mathbf{a}}_{\bullet,n+1}$ is the relative output of the economy as given by Leontieff's model.

Table 13
Input–output matrix

	Agriculture	Manufacturing	Services	Other	Final demand	Total output
Agriculture	10	65	10	5	10	100
Manufacturing	40	25	35	75	25	200
Services	15	5	5	5	90	120
Other	15	10	50	50	100	225
Value added	20	95	20	90	0	
Total input	100	200	120	225	225	645

Table 14
The supermatrix for the input–output matrix

$W =$	0.1	0.325	0.083333333	0.022222222	0.044444444
	0.4	0.125	0.291666667	0.333333333	0.111111111
	0.15	0.025	0.041666667	0.022222222	0.4
	0.15	0.05	0.416666667	0.222222222	0.444444444
	0.2	0.475	0.166666667	0.4	0

Table 15
Limiting supermatrix of the input–output table

0.114942529	0.114942529	0.114942529	0.114942529	0.114942529
0.229885057	0.229885057	0.229885057	0.229885057	0.229885057
0.137931034	0.137931034	0.137931034	0.137931034	0.137931034
0.25862069	0.25862069	0.25862069	0.25862069	0.25862069
0.25862069	0.25862069	0.25862069	0.25862069	0.25862069

Example of an input–output matrix

Consider the following input–output matrix shown in Table 13.

The supermatrix corresponding to this input–output model is given in Table 14. And the limiting matrix $\lim_{k \rightarrow \infty} W^k$ is given in Table 15.

$$\bar{w}_n = \begin{pmatrix} 0.114942529 \\ 0.229885057 \\ 0.137931034 \\ 0.25862069 \end{pmatrix} \text{ and by normalizing to unity we can see that}$$

$$\bar{w}_n (1 - w_{n+1})^{-1} = \begin{pmatrix} 0.15503876 \\ 0.310077519 \\ 0.186046512 \\ 0.348837209 \end{pmatrix} = \frac{1}{645} \begin{pmatrix} 100 \\ 200 \\ 120 \\ 225 \end{pmatrix}$$

which coincides with the normalized values of the total output of the economy.

Table 16 displays the input–output table (supermatrix) of the economy of the Sudan in 1976 with eigenvector values as the estimates [9] and the actual values in parentheses as computed by the Nobel Laureate Lawrence Klein who participated in the study with his Wharton Forecasting Associates. The results of this fairly complex exercise using paired comparison judgments are generally close to those of the econometric forecasting model.

Desirability of drilling for oil in Alaska — the ANWR model

In late 2002 a study was done by Emanuel and Cefalu [11] to find out whether drilling for oil should be allowed in the Arctic National Wildlife Reserve (ANWR) in northern Alaska. Environmentalists, mostly living in the lower 48 US states have been blocking drilling in the region. In their ANWR study the alternatives were: Do Drill, Do Not Drill.

Table 16

Input–output table of the Sudanese economy (1976) by the Wharton Forecasting Associates

	Agriculture	Public utilities	Manufacturing and mining	Transportation and distribution	Construction	Services
Agriculture	.0079 (.00737)	0 (0)	.2331 (.21953)	.0008 (.00042)	.0699 (.06721)	0 (0)
Public utilities	.0009 (.00024)	0 (0)	.0130 (.01159)	.0075 (.00618)	0 (0)	.0033 (.00283)
Manufacturing and mining	.0041 (.00393)	0 (0)	0 (0)	.0089 (.00857)	.0379 (.04216)	.0037 (.00322)
Transportation and distribution	.0691 (.06993)	.1694 (.145360)	.1281 (.12574)	0 (0)	.1115 (.09879)	.0153 (.00641)
Construction	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	.0546 (.05402)
Services	0 (0)	.0117 (.01030)	.0224 (.02549)	.0224 (.02422)	.0039 (.00520)	.0004 (.000210)

The study results were compared against a poll of Alaskan residents asking the question: “Do you think we should drill, or not drill in ANWR?”

ANWR-Arctic National Wildlife Refuge covers 19 million acres on the Northern coast of Alaska north of the Arctic Circle and 1300 miles south of the North Pole. The consensus of the geologic community is that the Coastal Plain of ANWR represents the highest petroleum potential onshore area yet to be explored in North America. If explored, it is estimated that it will take 15 years or more before oil and gas will reach the market. Legislation was passed in the 1980's that created a majority of the National Parks in Alaska and expanded ANWR to its current size. The Reagan Administration was ready to drill but was derailed by the Exxon Valdez catastrophe. The first Bush Administration likewise wanted to drill, but was unsuccessful. The Clinton Administration designated it as a protected area and it has been that way ever since then. The second Bush Administration, in response to ongoing Middle East violence and the 9/11 terrorist attacks, sees drilling in ANWR as vital not only for economic but national security reasons. Several environmental groups consider ANWR a great American natural treasure and one of the last places on the earth where an intact expanse of arctic and sub-arctic lands remain protected. They feel the habitat, the wildlife, and the culture need to be protected from the exploration of gas and oil. The top-level main network is shown in Fig. 11.

Connected to each of the BOCR merit nodes is the subnet for Benefits that contains a hierarchy of these control criteria: Economic, Political, Social. The control criteria in Fig. 12 were pairwise compared for importance and their values derived.

The subnet for benefits containing a hierarchy of control criteria is shown in Fig. 12. Connected to each control criterion node under benefits is a decision network containing the alternatives of the problem. The decision subnet for the economic control criterion, the most important of the benefits control criteria, is shown in Fig. 13. The final results are shown in Fig. 14 where the subnet results are combined in the main network using the formula $bB + oO - cC - rR$ where B, O, C, and R are the vectors of values for the decision alternatives from the decision subnets, and b, o, c, and r are the priorities of the benefits, opportunities, costs and risks nodes in the top-level network. The priorities of these nodes are obtained by rating them as to the impact on the strategic criteria of their top priority alternatives.

The strategic criteria here are General Public Opinion, International Politics, and Amount of Oil. They are first pairwise compared for importance, then used to rate the importance of the top rated alternative for each of the Benefits, Opportunities, Costs and Risks, called the merits of the decision, as shown in the table. The ratings categories themselves are pairwise compared to establish priorities for High, Medium and Low. To select the appropriate rating, one must keep in mind the highest priority alternative under the merit being evaluated. For example under risks: Do Not Drill is the highest priority alternative meaning that it is the most risky. One has to keep in mind for example, that the Risk merit under General Public Opinion is evaluated as low. The results of rating the merits are shown in Fig. 14 in which the priority of the Benefits is .407, closely followed by Opportunities at .364. In this decision, Costs are found to be unimportant, and Risks are nearly half as important as Benefits or Opportunities. These values for the merits nodes are used to weight the values for the alternatives as determined in the subnets they control to give the overall results for the alternatives shown in Fig. 15.

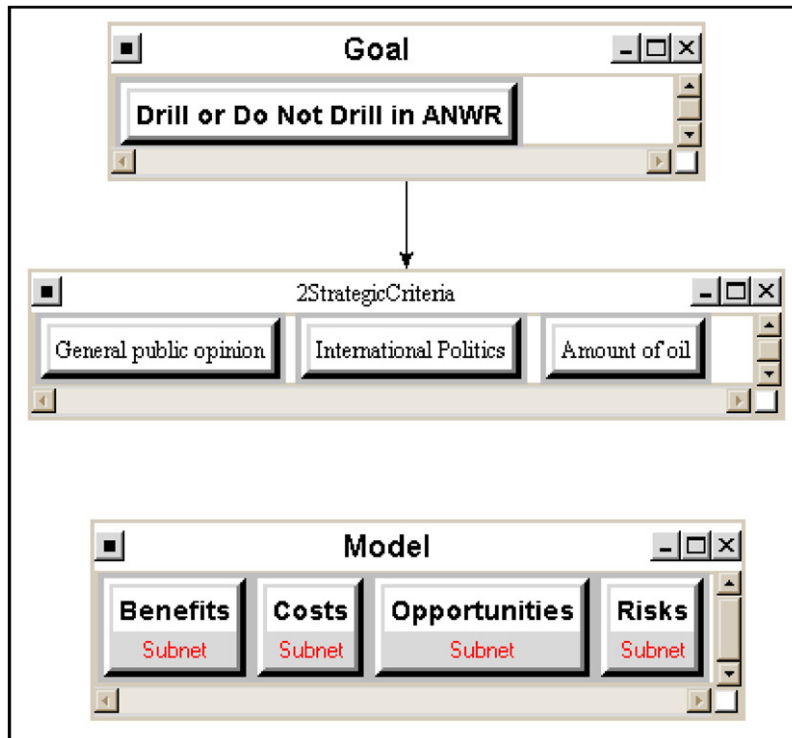


Fig. 11. The top-level network of the ANWR decision model.

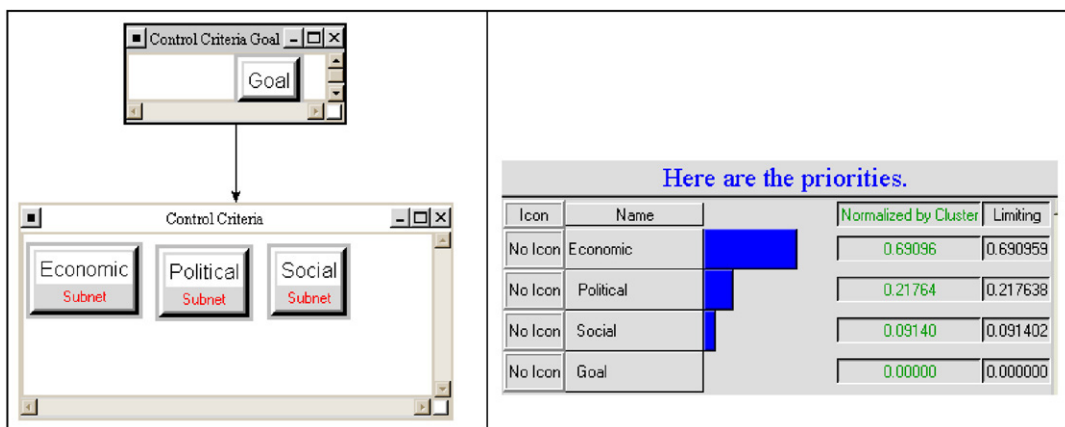


Fig. 12. Benefits subnet of control criteria and their priorities.

The final results of the ANWR study on whether or not to drill for oil in Alaska show that “Drill” gets 77.6%, and “Do Not Drill” gets 22.3% as shown in the Normals column of Fig. 15. This agrees quite well with a poll done in Alaska among residents as to whether or not to drill for oil.

A recent poll (2002) conducted among native Alaskans found that they supported opening ANWR to oil and gas exploration 75% to 19% with 6% undecided. The question asked was “Do you believe oil and gas exploration should or should not be allowed within the ANWR Coastal Plain?” Assigning the 6% equally yields 78% to 22%, the results of the poll are very close to the results of the model.

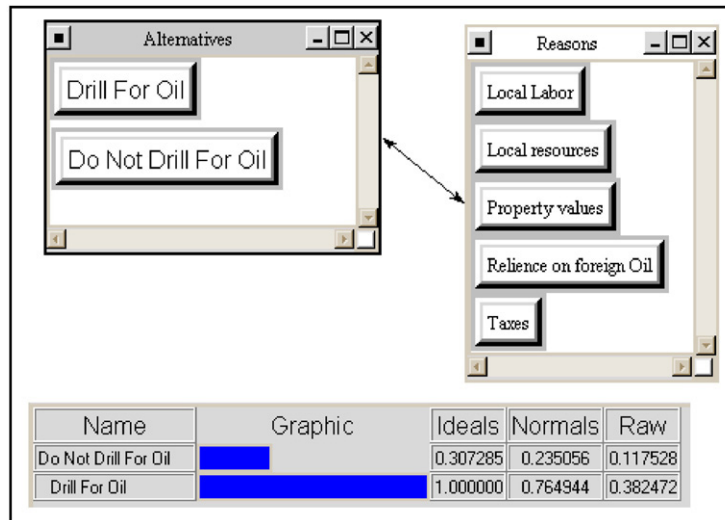


Fig. 13. The decision subnet under economic benefits and its synthesized outcomes.

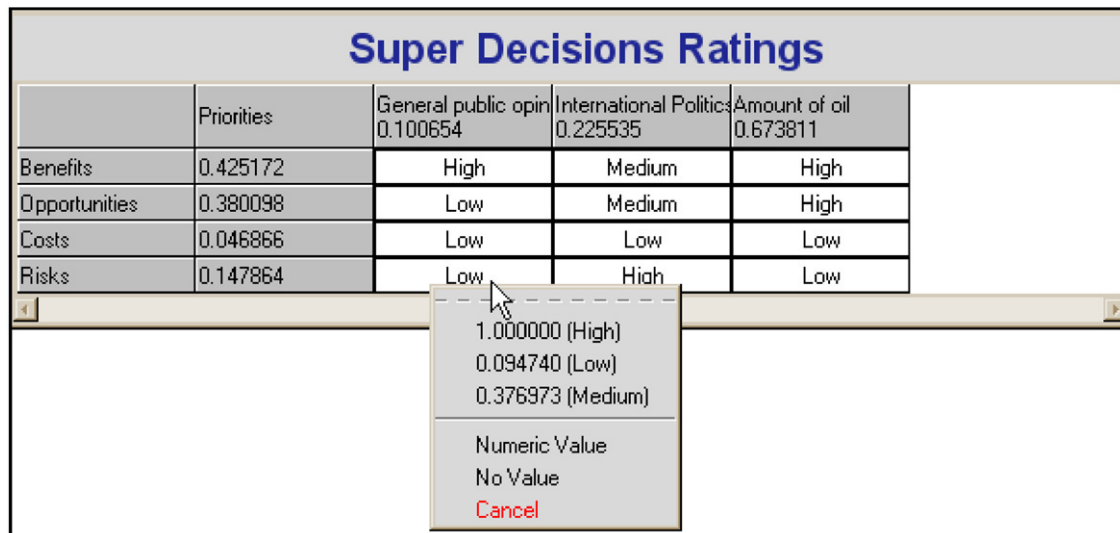


Fig. 14. BOCR merits by rating the impact of the top alternative of each on the strategic criteria.

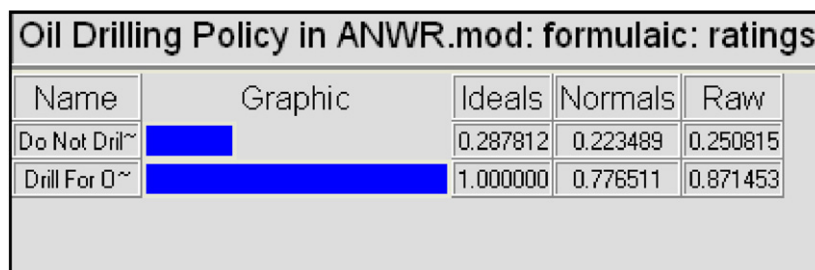


Fig. 15. Final results of the ANWR study.

5. Conclusions

The validation examples shown in this paper were chosen for their relative simplicity and importance from among the many that have been done. It goes without saying that someone who uses the process without adequate knowledge and understanding might not get an answer that can be validated against observation. The reason is that descriptive applications need a modicum of informed knowledge about the world and about the issue being examined. Otherwise the person doing the model may not be able to successfully structure the model and compare all the elements included in the model with sufficient accuracy. That is why group decision making tends to give better results because of the broader knowledge available and also because of the possibility of debates that may arise and change people's understanding — such changes in understanding that lead to improvements in the model or judgments are an important aspect of this process that can lead to improved decisions.

References

- [1] L. LeShan, H. Margenau, *Einstein's Space and Van Gogh's Sky*, MacMillan Publishing Company, New York, 1982.
- [2] D. Buede, D.T. Maxwell, Rank Disagreement: A comparison of multi-criteria methodologies, *Journal of Multi-Criteria Decision Analysis* 4 (1995) 1–21.
- [3] T.L. Saaty, M.W. Khouja, A measure of world influence, *Journal of Peace Science* (Spring) (1976).
- [4] T.L. Saaty, L.G. Vargas, Hierarchical analysis of behavior in competition: Prediction in chess, *Behavioral Sciences* (1980).
- [5] T.L. Saaty, L.G. Vargas, *Prediction, Projection and Forecasting*, Kluwer Academic, Boston, 1991.
- [6] A. Blair, R. Nachtmann, J. Olson, T.L. Saaty, Forecasting foreign exchange rates: An expert judgment approach, *Socio-Economic Planning Sciences* 21 (6) (1987) 363–369.
- [7] T.L. Saaty, Y. Cho, The decision by the US Congress on China's trade status: A multicriteria analysis, *Socio-Economic Planning Sciences* 35 (2001) 243–252.
- [8] L.G. Vargas, Personal communication. Unpublished.
- [9] T.L. Saaty, L.G. Vargas, Estimating technological coefficients by the analytic hierarchy process, *Socio-Economic Planning Sciences* 13 (1979) 333–336.
- [10] A. Blair, R. Nachtmann, T.L. Saaty, R. Whitaker, Forecasting the resurgence of the US economy in 2001: An expert judgment approach, *Socio-Economic Planning Sciences* 36 (2001) 77–91.
- [11] J. Emanuel, P. Cefalu, Drilling for oil in the Arctic National Wildlife Refuge, project in a graduate class: *Decision Making in Complex Environments*, Katz Graduate School of Business, University of Pittsburgh, 2002.